An entry–exit law for dead-neuron resurrection: the smoothed-ReLU limit of training dynamics at an activation boundary, and its confirmation in a small MLP and a one-layer transformer

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Abstract

A neuron "dies" when its pre-activation margin crosses zero on all data; for a hard ReLU the dead unit has exactly zero gradient and never returns. We ask what governs death and resurrection in the smoothed activations used in practice. For shallow softplus networks under gradient flow we prove that, in the sharp limit $\beta \to \infty$ of the activation stiffness, death and resurrection are governed by a hybrid system *with memory*: the dead unit's parameters freeze and reactivation occurs exactly when the signed time integral of the excluded residual returns to zero — the classical entry–exit function — with a universal first-order release coefficient; all constants are verified on one two-neuron instance. The two limit orders do not commute: the discrete ReLU flow freezes dead units forever, the smoothed flow revives them by the area law. We then track the law upward through a pre-registered experimental ladder: a reduced stochastic-schedule model is falsified; two instrument designs are recorded as negatives (a calibrated null fires); and a confirmatory MLP run (400 runs, $T = 0$) reproduces resurrection at $p = 0.50$ with the residual area opening before 13/16 returns, hard ReLU giving zero, and a monotone channel staircase in β. A pre-registered one-layer-transformer confirmatory (800 runs, $T = 0$) then closes the ladder: the theorem's class transfers to the FFN's own channel (softplus-β3 with attention frozen, $p_{\text{sust}} = 0.440\,[0.300, 0.587]$, drive area opening before the return in 21/22); a *universal upstream channel* adds $\approx +0.20$ in all four activation arms — including hard ReLU, whose own channel is exactly zero; and the GELU sign-flip barrier is *consistent with*, not established by, the counts, which are confounded with injection depth, while its sign signature separates deep from shallow in 92% against 4%.

1 Introduction

Write $\Phi : \mathcal{W} \to \mathcal{F}$ for the parametrization of a network and $J = D\Phi$ for its Jacobian. A neuron whose pre-activation margin is negative on every datum contributes an identically zero column to J: it sits on a stratum of the singular locus where J drops rank. Under a hard ReLU the gradient of the loss in the directions of that unit is exactly zero, so the stratum is *absorbing* — a dead unit stays dead. This is the mechanism behind the empirically documented dormant-neuron and loss-of-plasticity phenomena [9, 10, 11]. The question of this paper is what replaces the absorbing stratum when the activation is *smoothed*, as it is in every softplus/GELU network actually trained.

Our answer is an exact gating law and its numerical arc. The contributions are:

- **A theorem (§3, Thm. 1).** In the sharp limit $\beta \to \infty$ of a shallow softplus network under gradient flow, the death/resurrection of a neuron is the entry–exit problem of a

slow–fast system: the dead unit freezes and reactivates when $\int R\,dt = 0$, where R is the residual excluding that unit. The uniform rate is $O(\log\beta/\beta)$, the release time carries a *universal* first-order coefficient $-1/(cR(t_{\text{exit}}))$, and the law is independent of the smoothing within the monotone-sigmoid class (softplus, erf); GELU is excluded because its derivative is non-monotone. Six lemmas, each with an executable verification, are summarized in §3 and stated in Appendix A; Figs. 1–2 show the mechanism and the predicted constants.

- **Non-commutation (§4).** The two limit orders (smooth-then-sharp vs. the pointwise generalized-gradient flow) do not commute. The field-first order is the frozen branch of Eberle, Jentzen, Riekert and Weiss [8]; our flow-first limit resurrects. This is the exact mathematical content of "dead ReLU units stay dead, smooth units come back."

- **A pre-registered experimental ladder (§5).** Organized by prediction, with the null controls and the negatives shown: a reduced deterministic-schedule model *falsified* under SGLD (§5.1); two instrument designs recorded as *negatives* because a calibrated null fired (§5.2); the mechanism first observed in an MLP pressure pilot (§5.3); and a *confirmatory* MLP run — 400 runs, area law reproduced with confidence intervals, hard ReLU as the zero-gradient control, monotone staircase in β (§5.4, Fig. 4); and a *confirmatory* one-layer transformer — 800 runs, the theorem's class transferring in the FFN's own channel, a universal upstream channel of $\approx +0.20$, and the GELU barrier reported as consistent-with-barrier under an explicit depth confound (§5.5), whose scoping had already exposed both new mechanisms.

The paper is deliberately honest about scope: the theorem is one two-neuron instance; the confirmatories are two small architectures at temperature zero; the stochastic gate is open and, where we probed it, it *breaks* the deterministic schedule (§5.1).

2 Setup and notation

Data $(x_i, y_i)_{i=1}^n$ in general position; shallow network $f_\theta(x) = \sum_k g_\beta(w_k x + b_k)$ with softplus $g_\beta(z) = \beta^{-1}\log(1+e^{\beta z})$ (general monotone sigmoids in §3); quadratic loss $L = \frac{1}{2n}\sum_i (f_\theta(x_i) - y_i)^2$; gradient flow $\dot\theta = -\nabla L$. The parameter β is the *activation stiffness*: as $\beta \to \infty$, $g_\beta \to$ ReLU and $g'_\beta \to \mathbf{1}_{\{z>0\}}$.

Fix one neuron A and one datum j. Its *margin* is $z = \theta_A \cdot (x_j, 1)$; write $c = \|(x_j,1)\|^2/n$ and let

$$R(t) = y_j - f_{-A}\big(x_j; \theta_B(t)\big)$$

be the *residual excluding* A (θ_B collects the other parameters). The *hybrid reference* θ^{hyb} runs the full ReLU flow until the first time t_{hit} with $z = 0$, then freezes θ_A and runs θ_B alone; set R^{hyb} accordingly and $A^{\text{hyb}}(t) = \int_{t_{\text{hit}}}^t R^{\text{hyb}}$ for its signed area. The death of A is exactly the rank drop of J on the stratum $\{z \le 0\}$; the question is the fate of the trajectory on that stratum.

Hypotheses. **(H1)** $R^{\text{hyb}}(t_{\text{hit}}) < 0$ (transversal death). **(H2)** A^{hyb} has a first zero at $t_{\text{exit}} < T$ with $R^{\text{hyb}}(t_{\text{exit}}) > 0$ and $A^{\text{hyb}} < 0$ on $(t_{\text{hit}}, t_{\text{exit}})$. **(H3)** the other margins of A stay $\le -\delta_1 < 0$ on the episode. **(H4)** no other switching event occurs in $[0, T]$.

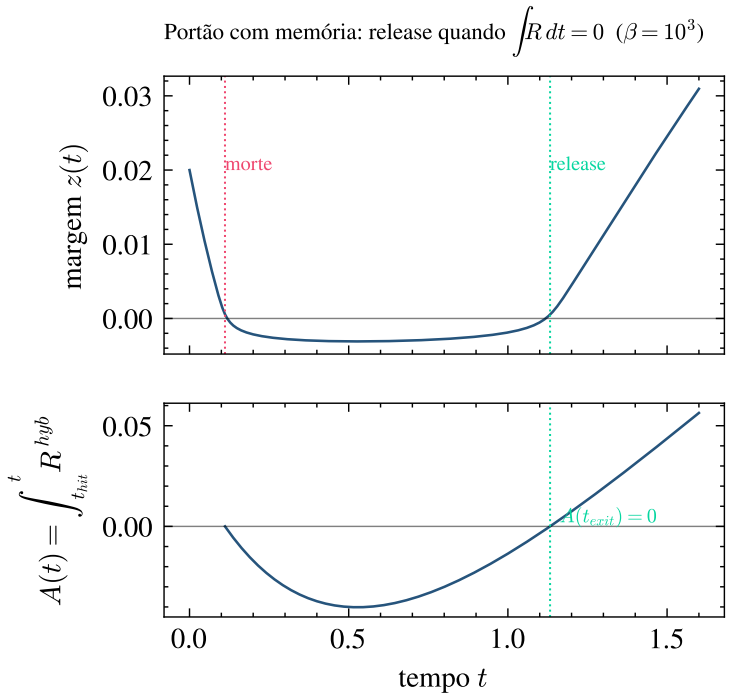


Figure 1: The memory gate at $\beta = 10^3$ (coupled system, reference instance). Top: the margin $z(t)$ dies at t_{hit}, hugs the sticking layer of depth $\sim \beta^{-1}\log\beta$, and releases. Bottom: the hybrid area $A(t) = \int_{t_{\text{hit}}}^{t} R^{\text{hyb}}$; release coincides with $A = 0$ up to $O(\beta^{-1})$ (Thm. 1). Deterministic Euler integration ($dt = 10^{-5}$); uncertainty is discretization only. Script `t2-prova/figuras_draft.py` (committed).

3 The entry–exit theorem

Definition 1 (hybrid limit)**.** *θ^{∞} equals: the ReLU flow on $[0, t_{\text{hit}}]$; on $[t_{\text{hit}}, t_{\text{exit}}]$, $\theta_A \equiv \theta_A(t_{\text{hit}})$ (frozen) with θ_B the B-only flow; on $[t_{\text{exit}}, T]$, the ReLU flow restarted at $(\theta_A(t_{\text{hit}}), \theta_B^{\text{hyb}}(t_{\text{exit}}))$.*

Theorem 1 ($T2'$-min)**.** *Under (H1)–(H4), for every $\delta > 0$,*

$$\sup_{t\in[0,T]\setminus(B_\delta(t_{\text{hit}})\cup B_\delta(t_{\text{exit}}))} \|\theta^{\beta}(t) - \theta^{\infty}(t)\| = O_\delta\Big(\tfrac{\log\beta}{\beta}\Big), \qquad t^{\beta}_{\text{release}} = t_{\text{exit}} - \frac{1}{c\,R^{\text{hyb}}(t_{\text{exit}})}\cdot\frac{1}{\beta} + o(\beta^{-1}),$$

where $t^{\beta}_{\text{release}}$ is the first return of the margin z to 0 after death. The $\log\beta$ in the sup is sharp and is carried solely by the normal coordinate of θ_A inside the episode (the sticking depth $z \sim -\beta^{-1}\log\beta$); θ_B obeys $O(\log\beta/\beta^2)$ and the tangential components of θ_A are exponentially frozen. The first-order release coefficient is universal*: in the product (entry delay × area cut) the arrival speed cancels, leaving the reciprocal exit speed — each corner contributes the reciprocal of its transversal speed. The remainder is $o(\beta^{-1})$ as proved. The limit is that of the regularized flow; discrete ReLU gradient descent freezes the dead neuron forever (the limits do not commute, §4).*

Proof outline. The episode reduces to a scalar entry–exit problem through six lemmas (Appendix A); each has a committed executable symbolic/numeric verification (`passo0--6`,

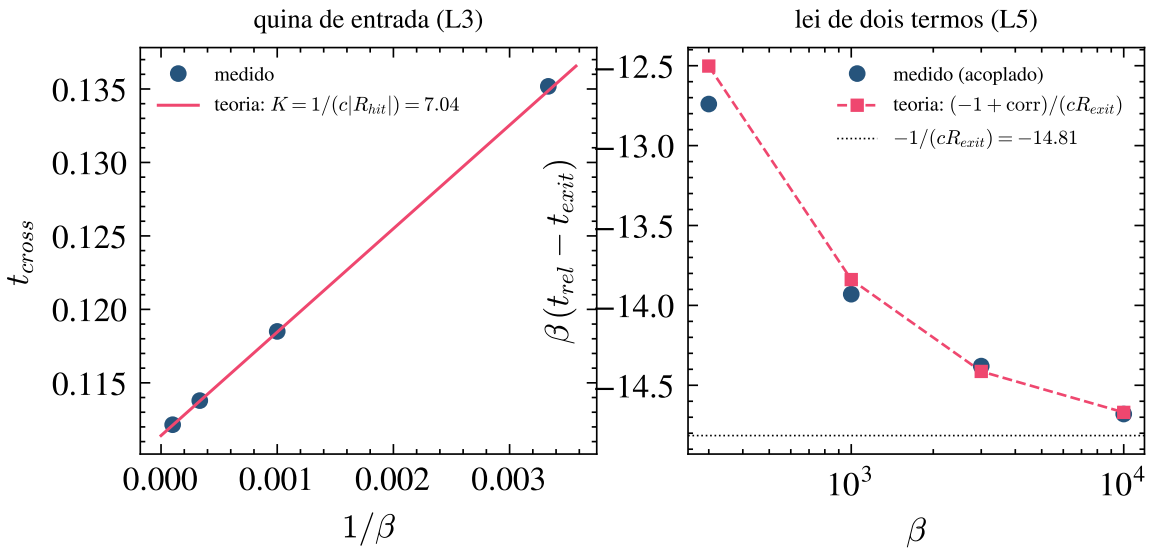


Figure 2: Predicted constants vs. measurements (deterministic runs; uncertainty is discretization). Left: entry-corner crossing times against $1/\beta$; the line is the *proved* slope $K = 1/(c|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|) = 7.04$ (Lem. 3), not a fit. Right: the two-term release law (Lem. 5, squares) against the coupled measurements (circles) over $\beta \in [3 \cdot 10^2, 10^4]$; dotted line: the universal first-order coefficient $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})) = -14.81$. Script `t2-prova/figuras_draft.py`.

`prova_L1--L6`). In the gauge $w = e^{\beta z}$, fast time $\tau = \beta t$, the margin obeys the quadratic normal form of De Maesschalck–Schecter [1] with $\varepsilon = \beta^{-1}$ (Lem. 1); with $u = 1/w$ and $\Phi(u) = \log u + u$ one has the exact additive quadrature $d\Phi/dt = -c\beta R + c\log(1 + w)$, so the layer is held by the *area*, not by the sign of R (Lem. 4): $u(t) = c\beta|A(t)|(1 + o(1))$ returns to $O(1)$ at $t_{\mathrm{exit}} + O(\beta^{-1})$. The B-subsystem is affine and, conditionally on the layer being maintained, tracks the hybrid at $O(\log\beta/\beta^2)$ (Lem. 2). A coupled continuation (nonempty–open–closed) closes the feedback loop and yields the two-term release law (Lem. 5); the entry and exit corners contribute mirrored $1/\beta$ terms whose arrival-speed factor cancels (Lems. 3, 6). The dead face carries an identically zero field on one side — a degeneracy outside generic piecewise-smooth exit theory [4, 3] and of the exponentially flat type of Kristiansen [2]; the canonical Sotomayor–Teixeira regularization [6, 7] requires both one-sided normal components nonzero and so does not cover it. The coupling of the forcing to the flow (the residual is generated by the very trajectory it gates) has, to our knowledge, no analogue in the entry–exit literature [5].

Numerical verification (one instance). $X = (0.5, 1.5)$, $Y = (0.5, 1.0)$, $\theta_A(0) = (-0.5, 0.27)$, $\theta_B(0) = (1.144, 0.261)$; $t_{\mathrm{hit}} = 0.1114$, $t_{\mathrm{exit}} = 1.1325$, $\delta_1 = 0.508$. Every empirically measured constant is predicted by the proofs: entry coefficient $K_{\mathrm{teo}} = 7.04$ vs 7.13 fitted (1.2%); the coupled two-term law $(-1 + \mathrm{corr})/(cR_{\mathrm{exit}})$ reproduces the fine-grid coupled sweep $-12.74/-13.93/-14.38/-14.68$ at 1.9%/0.7%/0.2%/0.1% for $\beta = 3 \cdot 10^2 \ldots 10^4$ (universal coefficient $1/(cR_{\mathrm{exit}}) = 14.815$; against -14.68 the leading coefficient alone is at 0.9%); tangential drift at machine zero. All numerics are this single two-neuron instance; breadth is future work.

Regularization independence. The release criterion depends only on $\int R\,dt = 0$, not on the sigmoid: for the erf class ($g'_\beta = \Phi_{\mathrm{gauss}}(\beta z)$) the coupled system releases at the same t_{exit} with

$O(\beta^{-1})$ gap (verified). GELU itself, $z\,\Phi_{\text{gauss}}(z)$, has a non-monotone derivative and is *not* a member of the family for any β; §5.5 shows this is not a technicality but a distinct mechanism.

4 Non-commutation and the plasticity bridge

The theorem is the flow-first limit: take $\beta \to \infty$ of trajectories. The *field-first* order — flow the pointwise-limit generalized gradient — instead freezes degenerate neurons forever; this is the rigorous branch of Eberle, Jentzen, Riekert and Weiss [8], whose selected solution has a non-decreasing degenerate set (the convention $\mathrm{ReLU}'(0) = 0$ and the no-revival mechanism are the same, though their degenerate stratum is the fully-zero one, not literally our functionally dead neuron). The two orders do not commute: field-first freezes, flow-first resurrects by the area law.

In machine-learning terms this non-commutation is the statement behind an empirical trend: discrete ReLU units that die stay dead — the dormant-neuron and loss-of-plasticity phenomena documented across deep and reinforcement learning [9, 10, 11], mitigated in practice by recycling dead units or by migrating to smooth activations — whereas the smoothed flow resurrects them by the area law. The channel that reopens a smoothed unit is $e^{\beta z}$, which closes as $\beta \to \infty$; §5.4 measures the resulting staircase directly, and the most alive arm ($\beta = 3$) is the smoothness of the GELU family.

5 Experiments

The claims are tested by prediction, each with its null control, and the failed designs are shown rather than hidden. Every confirmatory analysis was frozen by a pre-registration commit before the run; hypotheses that fell were archived, not rewritten. Commits: piloto-pressure `b48b527`; rate pilot `995a87c`; 4a.3 pre-registration `a8f8f8c`; 4a.3 confirmatory `5a961cb`; transformer (etapa-5) pre-registration `1b45498`, confirmatory result `4051e7f`.

5.1 The deterministic schedule is falsified under SGLD

Before any confirmatory claim, we tested whether the deterministic entry–exit schedule survives noise. A reduced "race" model (pre-registered, commit `190d94a`, two adversarial-review rounds) predicts the capture (resurrection) probability of a dead unit from the mean residual schedule; the falsifying run (2700 runs) rejected it. The predicted-zero window was violated decisively: 139 late captures occurred where the model gives exactly zero, and the bin $[0, 0.1)$ saw 150 captures against 1.4 predicted ($z = +131$). Crucially the model is well calibrated where it says capture is *likely* (the $(0.5, 0.9)$ bin, $z = -0.26$); the entire error is on the "impossible" side. The theorem-part of the model held: the observed censoring (2–7%) matched the Brownian-cone prediction in all nine cells — so the failure is of the *hypothesis* (a deterministic schedule), not of the instrument.

An erratum records the next step honestly: the natural fix (a stochastic Ornstein–Uhlenbeck schedule, no free parameter) predicts p(capture) nearly identical to the deterministic one (differences ≤ 0.013) and a late-capture rate of 0.1–1.5% against the $\sim 13.8\%$ observed — insufficient by an order of magnitude. That model was therefore refuted at the predictor stage and *not run*; the leading candidate became the back-reaction of the reactivating unit on the shared residual. The stochastic gate remains open and is out of scope for the theorem, whose regime is deterministic gradient flow.

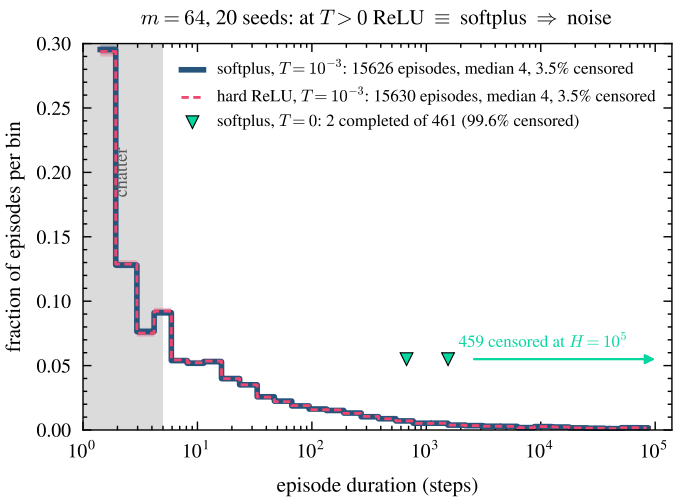


Figure 3: Why etapa-4a is an instrument negative (post-hoc diagnostic, $m = 64$, 20 seeds). Episode-duration densities: at $T > 0$, hard ReLU $\equiv$ softplus (median ~ 4 steps, shaded chatter band) — a signature of noise, not of the zero-gradient theorem; the real physics is the $T = 0$ arm ($\sim 10^3$ steps, 99.6% censored). Script `etapa4-mlp-populacao/figuras_diagnostico.py`.

5.2 Two instrument designs recorded as negatives

The first attempt to see the law in a real MLP failed on the *instrument*, and the failure was caught by a pre-registered null (etapa-4a, commit `6411ac8`, 600 runs, MLP $m \in \{16, 64, 256\}$, SGLD $\eta = 3 \cdot 10^{-4}$). The pure-diffusion arm passed the calibrated test ($m = 16$, $p = 0.002$); by the pre-registered rule this means the ruler is biased and *no* scientific verdict on the substantive predictions may be claimed — the run is **invalid**. Diagnosis (Fig. 3): the ruler was measuring boundary *chatter*, not the theorem's physical death. 59% of episodes lasted ≤ 5 steps; hard ReLU "resurrected" at the same rate and with the same duration distribution as softplus at $T > 0$ — impossible by gradient (a dead ReLU unit has exactly zero gradient), hence noise threshold-crossings. The real physics lives at $T \to 0$: the only genuinely long episodes (median ~ 1110 steps) sit in the $T = 0$ arm, 99.6% censored at the 10^5-step horizon. The calibrated null held false-positive rate near α in i.i.d. worlds, but the real world has per-neuron heterogeneity (non-i.i.d.).

A redesign (etapa-4a.2) was itself rejected at the design stage by the same process (commit `a413b19`): at $T = 10^{-5}$ the ReLU/softplus ratio stayed ≈ 0.95 after debounce (gradient starvation at $\beta = 300$ masked the effect), and the pooled null broke α by 30–92$\times$ in heterogeneous/autocorrelated worlds. The inheritance was explicit: a hysteresis/dwell ruler, a per-neuron stratified null, and a move to the deterministic regime $T = 0$ with a much longer horizon. These negatives falsify neither the area law nor the toy calibration; they falsify two rulers, and they are what forced the working design.

5.3 The mechanism, observed

Moving to $T = 0$ and a sub-parametrized regime (pressure: a plateau ~ 0.15 on the rich target $y = \sin(2\pi x) + 0.3\sin(5\pi x)$, $n = 16$, $\eta = 10^{-3}$, $H = 2 \cdot 10^5$; 64 runs, commit `b48b527`, exploratory) made the mechanism visible. Hard ReLU gave a *structural zero*: 0 resurrections out of 25 deaths

arm	dyn. deaths	λ/run [boot]	returns	p(ret | death) CP	ordered/returns
sp-β3	35	0.350 [0.220,0.500]	16	0.457 [0.288,0.634]	13/16
sp-β10	27	0.270 [0.160,0.390]	10	0.370 [0.194,0.576]	6/10
sp-β300	19	0.190 [0.100,0.290]	4	0.211 [0.061,0.456]	3/4
ReLU	17	0.170 [0.090,0.260]	0	0.000 [0.000,0.195]	0/0

Table 1: Confirmatory 4a.3 (100 runs/arm, $T = 0$). ReLU gives a structural zero (validity control: any ground-truth return here would invalidate the run). Clopper–Pearson 95% intervals; bootstrap clustered by seed.

in 16 seeds, the frozen (grad $\equiv 0$) branch exact in a real network at $T = 0$. Softplus resurrected by mechanism: at $\beta = 10$ a unit died at step 781 and returned at 139,179; at $\beta = 300$ one died at 3315 and returned at 152,977 (it had parked shallow enough that the $e^{\beta z}$ channel did not vanish). Both dynamical resurrections showed the ordered triple $t_{\text{death}} < t_{\text{area}} < t_{\text{res}}$ (area law 2/2). The gate→return gap is the finite-β rise time ($\sim |z_{\text{park}}|/\text{drift}$), not the asymptotic $O(1/\beta)$ correction. The bottleneck was quantified as the event rate (~ 0.1/run) — which set the entire sizing of the confirmatory.

5.4 Confirmatory: the area law in a real MLP

The confirmatory (etapa-4a.3) was pre-registered at commit `a8f8f8c` after two more adversarial rounds, with every design number derived programmatically from a 240-run rate pilot (commit `995a87c`); the run is commit `5a961cb`. Grade: {sp-β3 (primary), sp-β10, sp-β300, ReLU} $\times$ $m \in \{4, 6\} \times 50$ virgin seeds (1900–1949) = 400 runs; $T = 0$, $H = 5 \cdot 10^5$; eligible window $[0, 307{,}536]$ ($H - \tau$, τ the $q_{0.9}$ return time). The instrument is a single ground-truth clock with bit-exact parity to the pilot motor; the area of the leader-datum residual is accumulated on that clock, so temporal ordering is structural and the only empirical failure mode of the area test is a *return without area*.

The pre-registered verdicts (Table 1, Fig. 4):

- **C1 (existence with rate, primary sp-β3): reproduces.** $N_{\text{elig}} = 30$ ($n^* = 10$), 15 eligible returns, $p(\text{ret} \mid \text{eligible}) = 0.500$ $[0.313, 0.687]$, well above the kill threshold 0.076.
- **C2 (the area law, the heart, sp-β3): sustains.** The leader-drive area opens before the return in 13/16 dynamical resurrections (lower Clopper–Pearson bound $0.544 > 0.5$, ≥ 6 returns). The 3 returns-without-area are real and counted against the law (returns driven off the leader datum — an object of study, not hidden).
- **C4 (channel staircase, estimation, no kill): monotone in β.** p(ret | death) falls $0.457 > 0.370 > 0.211 > 0.000$ from $\beta = 3$ to hard ReLU, with intervals — the activation smoothness controls the return channel, and $\beta = 3$ (the smoothness of the GELU family) is the most alive. This is the direct bridge to the plasticity practice of §4.
- **Validity: clean.** ReLU 0/17 dynamical returns — the non-commutation of Thm. 1 measured with $N = 100$.

The honest limits of the confirmatory: $T = 0$ (deterministic), one small architecture, a 1D rich target, $m \in \{4, 6\}$. The stochastic regime is a paper of its own (§5.1). C1's kill is intentionally near-unreachable at this N (a consequence of the frozen $\frac{1}{2}\times$ pilot-lower-bound rule), so in practice C2 is the deciding test; this was signalled in the pre-registration, not discovered after.

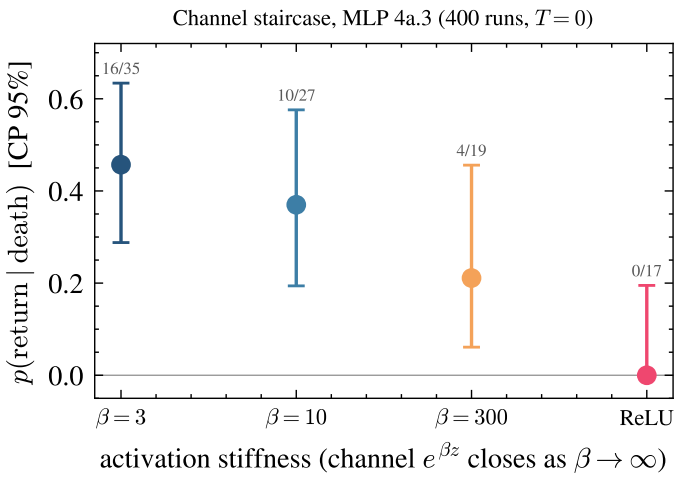


Figure 4: The channel staircase (C4, confirmatory 4a.3, 400 runs, $T = 0$). $p(\text{return} \mid \text{death})$ is monotone in the activation stiffness and reaches a structural zero at hard ReLU (grad $\equiv 0$); the smooth channel $e^{\beta z}$ closes as $\beta \to \infty$. Fractions above each point are returns/deaths; error bars are Clopper–Pearson 95%. Script `figuras_paper.py`.

5.5 Confirmatory: two new mechanisms in a one-layer transformer

The same question in a one-layer transformer (FFN over an attended token) does *not* reduce to the MLP, and the scoping (exploratory, commit `d4e4c35`) exposed why, with two facts that reshape the confirmatory:

- **S1 — a GELU sign-flip barrier.** $\text{GELU}'(z)$ changes sign at $z^* = -0.7518$ (verified analytically): for $z < z^*$ it is negative, so a deep-dead unit is pushed *away* from the return by its own gradient. Self-resurrection is possible only in the shallow band $(-0.752, 0)$. The MLP's blind area port (which omits g') gets the *sign* wrong on 12/12 GELU episodes; the correct instrument uses the slope-weighted area A_{drive}. GELU is therefore genuinely outside the monotone-sigmoid class of Thm. 1, not a technicality.

- **S2 — an upstream channel.** The FFN input $h = e + \text{attn}(e)$ is non-stationary, so a hard-ReLU unit with own-gradient identically zero can still revive because h moves. With attention frozen the control is exact: $\Delta\text{margin} = 0.000$, own_move $= 0$; with attention training, own_move $= 0$ still but the margin climbs $+0.69$ on average (up to $+1.32$). The MLP theorem "dead ReLU = terminal" holds only when the input is fixed; the transformer adds a channel the MLP does not have.

A rate pilot (160 runs, commit `2dba27d`) then fixed the design: natural FFN death is ≈ 0 (units in $d = 16$ input do not die on their own), so the confirmatory is necessarily injection-conditional; the non-stationary channel chatters, forcing a hysteresis instrument ($\delta_h \approx 0.0476$, calibrated from data); the return time is slow ($\tau \approx$ 60–66k), forcing $H = 2 \cdot 10^5$. The upstream channel was already clean in the pilot (Δ = train$-$frozen $= +6/+6/+6/+8$ returns; ReLU $0 \to 8$ with own_move $= 0$).

The transformer confirmatory (etapa-5, pre-registered commit `1b45498`; 800 runs, {sp-β3, sp-β10, GELU, ReLU} $\times$ {attn-train, attn-frozen}$\times$ injection $\{-0.3, -2.0\} \times 50$ seeds 2000–2049; $H = 2 \cdot 10^5$, $T = 0$, hysteresis $\delta_h = 0.0476$) ran to completion under the frozen analysis (result

commit `4051e7f`). Its five pre-registered estimands returned as follows; all counts are of *sustained* returns under the hysteresis ruler.

- **E-A (validity): valid.** Zero sustained returns in relu-frozen across 100 injections — 0/50 shallow and 0/50 deep. Own-gradient $\equiv 0$ with a fixed input is terminal, exactly as the theorem requires; any sustained return here would have invalidated the run.
- **E-B (upstream channel): reproduces in all four arms.** With $\Delta = p_{\text{sust}}(\text{train}) - p_{\text{sust}}(\text{frozen})$ and the bootstrap clustered by seed:

  $$\begin{array}{lll} \text{sp-}\beta 3 & \Delta = 0.210\,[0.120, 0.300] & (43/100 \text{ vs. } 22/100), \\ \text{sp-}\beta 10 & \Delta = 0.200\,[0.110, 0.280] & (29/100 \text{ vs. } 9/100), \\ \text{GELU} & \Delta = 0.220\,[0.120, 0.320] & (51/100 \text{ vs. } 29/100), \\ \text{ReLU} & \Delta = 0.200\,[0.120, 0.280] & (20/100 \text{ vs. } 0/100). \end{array}$$

  Every lower bound is above zero, and the effect is *additive and essentially the same*, $\approx +0.20$, across activations whose own channels differ by orders of magnitude. The ReLU arm is the clean case: its own channel is exactly zero, so the whole $0 \to 20/100$ is upstream. This is a mechanistic answer to why real networks recover dead units even under hard activations.
- **E-C (transfer of the theorem class): transfers.** sp-$\beta 3$ frozen with shallow injection — the FFN's own channel, isolated — gives 22/50 sustained returns, $p_{\text{sust}} = 0.440\,[0.300, 0.587]$ (Clopper–Pearson), against the pre-registered gate $n^* = 6$ and kill threshold 0.0062. The leader-drive area opens before the return in 21/22 of them. This is the direct analogue of the MLP's C2, and it holds.
- **E-D (GELU barrier): consistent-with-barrier (confounded with depth)** — the pre-registered three-way label, reported verbatim. The counts are gelu-frozen shallow 29/50, $0.580\,[0.432, 0.718]$, against deep 0/50, $0.000\,[0.000, 0.071]$; but the depth-matched control sp-$\beta 3$-frozen at -2.0 is *also* 0/50, so the counts alone do not separate the barrier from injection depth. What does separate them is the S1 signature itself: the sign inversion $A_{\text{drive}} \neq A_{\text{mlp}}$ occurs in $46/50 = 92\%$ of the deep injections against $2/50 = 4\%$ of the shallow ones. We therefore report the barrier as *consistent with* the data and visible in the sign channel — not as established by the count.
- **E-E (β staircase, estimation, no kill): monotone.** p_{sust} with attention frozen falls from $0.220\,[0.143, 0.314]$ at $\beta = 3$ to $0.090\,[0.042, 0.164]$ at $\beta = 10$; with attention training, from $0.430\,[0.331, 0.533]$ to $0.290\,[0.204, 0.389]$. The MLP staircase of §5.4 reappears in the transformer, in both the own and the upstream regimes: stiffening the activation closes the return channel.

Across the grid the temporal ordering predicted by the area law holds in the large majority of sustained returns — 21/22 (sp-$\beta 3$ frozen), 34/39 (sp-$\beta 3$ train), 26/27 (sp-$\beta 10$ train), 24/29 (GELU frozen) — so the drive area opens *before* the return, as in the MLP.

The scoping verdict survives the confirmatory and is itself a result: the MLP mechanism does not transfer directly to the GELU FFN — the sign-flip barrier obstructs self-resurrection — while the theorem's own class (softplus-β) does transfer, and the upstream channel is a genuinely new route that the MLP does not have. The confirmatory is injection-conditional by design (natural FFN death is ≈ 0 at $d = 16$), so it measures the fate of units placed dead, not the rate at which they die.

6 Limitations

We state them explicitly. **(i)** The theorem is proved for one two-neuron instance; every constant is verified there and breadth is future work. **(ii)** The regime is $T = 0$ deterministic gradient flow throughout the theorem and both confirmatories; the stochastic (SGLD) gate is open and, where probed, *falsifies* the deterministic schedule (§5.1) — late captures recur past the deterministic window at a rate the reduced stochastic model under-predicts, leaving the reactivating unit's back-reaction as the leading candidate driver. **(iii)** Scale: the confirmatories are small networks ($m \leq 6$ MLP units, $m_{\text{ffn}} = 4$ FFN units) with rich 1D targets, chosen so that a single unit can be tracked with a ground-truth clock. **(iv)** The transformer confirmatory is *injection-conditional*: natural FFN death is ≈ 0 at $d = 16$, so it measures the fate of units placed dead, not the rate at which they die; the design declares this and does not claim a death rate. **(v)** GELU is outside the monotone class, and our transformer verdict on its barrier is *consistent with*, not established: the deep-vs-shallow count is confounded with injection depth (the depth-matched softplus control is also 0/50), and only the sign signature separates the two (§5.5). **(vi)** The cone apex (corank ≥ 2), where weight decay produces genuine sliding toward the most singular stratum, is untouched.

7 Reproducibility

All runs are deterministic on CPU with per-run manifests (SHA256 + git + config); raw episode files and aggregate JSON are committed alongside the code. Pre-registration commits precede every confirmatory run in the repository history.

- **Theorem constants and figures.** `06-notas/t2-prova/`: lemma verifications `passo0--6.py`, `prova_L1--L6.py`; figures via `figuras_draft.py`.
- **MLP mechanism (pilot).** `etapa4-mlp-populacao/pilotos-4a2/pressao/`; commit `b48b527`.
- **MLP confirmatory (4a.3).** `.../pressao/taxa/`: pre-registration `a8f8f8c`, run `5a961cb`. `python run_4a3.py --workers 4` then `python analise_4a3.py` (reads `results-4a3/` + `config_4a3.json`). Rate pilot `995a87c`. Staircase figure: `06-notas/paper/figuras_paper.py`.
- **Transformer confirmatory (etapa-5).** `05-experimentos/etapa5-transformer/`: scoping `d4e4c35`, rate pilot `2dba27d`, confirmatory pre-registration `1b45498` (which freezes `config_e5.json`: $\delta_h = 0.0476$, 800 runs, seeds 2000–2049, $H = 2 \cdot 10^5$), result commit `4051e7f`.
  `python run_e5.py --workers 4` then `python analise_e5.py` (reads `results-e5/`, 800 episode files + manifest, and writes `resultado-e5.md`).
- **Negatives.** `05-experimentos/NEGATIVOS.md` and the etapa-4a note record the invalidated instrument (commit `6411ac8`) and the rejected redesign (`a413b19`); the falsified schedule is `p5-relu-morte-v2/` (commit `190d94a`).

A The six lemmas

Lemma 1 (normal form)**.** *In the gauge $w = e^{\beta z}$, fast time $\tau = \beta t$: $dw/d\tau = cR(t)\, w^2/(1+w)\,(1+E_\beta)$, $|E_\beta| \leq \log(1+w)/(\beta|R|)$ — the quadratic normal form of De Maesschalck–Schecter [1]*

with $\varepsilon = \beta^{-1}$. With $u = 1/w$ and $\Phi(u) = \log u + u$ one has the exact quadrature $d\Phi/dt = -c\beta R(t)(1 + E_\beta)$.

Lemma 2 (B-subsystem and feedback)**.** *Under (H4) the B-field is affine with Lipschitz constant $L_B = \lambda_{\max}(G_B)$ and R^{hyb} is affine in θ_B. Integral Grönwall consuming the bound of Lem. 1 gives, conditionally on the layer being maintained (supplied by Lem. 5), the refined rate $\sup \|\theta_B^\beta - \theta_B^{\mathrm{hyb}}\| = O(\log\beta/\beta^2)$ — one full order better than the naive $O(\beta^{-1})$: in the bulk $u = O(\beta|A|)$, so the coupling is $O(\beta^{-2})$ there and $O(\beta^{-1})$ only in the $O(\beta^{-1})$-wide corner windows. Verified: metric $\sup \cdot\, \beta^2/\log\beta$ stable in $[4.2, 5.7]$.*

Lemma 3 (entry corner)**.** *$\dot z^0(t_{\mathrm{hit}}^-) = cR^{\mathrm{hyb}}(t_{\mathrm{hit}}) < 0$ by (H1). The exact identity $(1-\sigma)/\sigma = e^{-s}$ collapses the crossing-time difference to an elementary integral: $t_{\mathrm{cross}}^\beta = t_{\mathrm{hit}} + \frac{1}{\beta v}(1 + O(\delta_z) + o(1))$ with $v = c|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|$, and the exponentially weighted approach integral is $\frac{3/2}{\beta v}(1 + o(1))$. Verified: $K_{\mathrm{teo}} = 7.04$ vs $K_{\mathrm{emp}} = 7.13$ (1.2%); weight constant to $< 1\%$.*

Lemma 4 (sticking and release, prescribed R)**.** *For prescribed R satisfying (H1)–(H2), with $A(t) = \int_{t_{\mathrm{hit}}}^t R$: the exact additive identity $d\Phi/dt = -c\beta R + c\log(1+w)$ holds (no division by R; the interior zero of R forced by (H1)–(H2) is harmless — the layer is held by the* area*, not the sign of R), and on compact subintervals of $(t_{\mathrm{hit}}, t_{\mathrm{exit}})$, $\Phi(u(t)) = -c\beta A(t)(1 + o(1))$, hence $u(t) = c\beta|A(t)|(1 + o(1))$ (depth $z \sim -\beta^{-1}\log\beta$), and u returns to $O(1)$ at $t_{\mathrm{exit}} + O(\beta^{-1})$ — no $\log\beta$ factor. Sub/super solutions from monotonicity in R give explicit constants; a perturbation δ of R shifts t_{exit} by $\le c\,\delta\,(t_{\mathrm{exit}} - t_{\mathrm{hit}})/|A'(t_{\mathrm{exit}})| + O(\delta^2)$.*

Lemma 5 (coupled bootstrap)**.** *Continuation with strict margins: the property "θ_B is $2M_2\log\beta/\beta^2$-close to the hybrid and the margin stays strictly in the layer after an $O(\beta^{-1})$ entry collar" is nonempty (Lem. 3), open (strict inequalities + continuity) and closed (the assumption returns the better constant M_2 via Lem. 2, and Lem. 4 with $\delta = O(\log\beta/\beta^2)$ keeps the layer up to $t_{\mathrm{exit}} - C_\varepsilon\log\beta/\beta$), with $M_2 = e^{L_B T} C_f C_4/c$ independent of the hypothesis for $\beta \ge \beta_0$; with the empirical $C_4 = 10.6$, $M_2 = 440.8$; the leading term of the theoretical C_4 is $2/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})) = 29.6$. The entry corner cuts initial area $\approx R^{\mathrm{hyb}}(t_{\mathrm{hit}})K/\beta$, advancing the exit by $1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})\beta)$ — the arrival speed cancels. Hence the two-term closed-loop law $\beta(t_{\mathrm{release}}^\beta - t_{\mathrm{exit}}) = (-1 + \mathrm{corr}(\beta))/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})) + o(1)$. Verified: the two-term law reproduces the entire coupled sweep at 1.9%/0.7%/0.2%/0.1% ($\beta = 3\cdot 10^2 \ldots 10^4$).*

Lemma 6 (exit corner; scalar reduction)**.** *Post-release the field is transversal outward ($cR^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$, (H2)); the mirrored corner identities of Lem. 3 apply with $v' = cR^{\mathrm{hyb}}(t_{\mathrm{exit}})$. Forward uniqueness of the restarted ReLU flow holds on the* activating branch*, selected by the limit itself (release at $t_{\mathrm{exit}} + O(\beta^{-1})$ is incompatible with the frozen Filippov branch); classical Grönwall glues on $[t_{\mathrm{release}}, T]$ at rate $O(\log\beta/\beta)$. Under (H3) the other data contribute $O(e^{-\beta\delta_1})$, the face-j field is rank one along $(x_j, 1)$, and the episode reduces to the scalar problem. Verified: gluing sup $= 1.6\cdot 10^{-3}/5.5\cdot 10^{-4}$ at $\beta = 10^3/3\cdot 10^3$.*

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